

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 1st Year 2nd Term Examination, 2016
Hum 1217
(Principle of Economics)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.

Section – A

1. (a) Explain Robbin's definition of Economics. Distinguish between Micro and Macro Economics. (10)
(b) What are the fundamental economic problems of an Economic an Economic organization? (10)
(c) "Scarcity, choice and opportunity cost goes together." Explain with the help of production possibilities frontier. (15)
2. (a) Define Law of demand. Explain why demand curve shifts. (15)
(b) There are 100 identical producers in the market, each with a supply function is given by $\Theta_{sx} = -40 + 20 P_x$ and 1000 identical individuals, each with a demand function is given by $\Theta_{dx} = 8 - P_x$. Determine equilibrium price and quantity. (10)
(c) What are the various factors that affect the supply curve to shift? (10)
3. (a) Define total revenue. Explain the behavior of total revenue on the basis of price-elasticity of demand. (15)
(b) The 'Grantho Shamvar' is a publisher of romantic novels – just stories of common people. A company hires an economist to determine the demand for its novel. After months of hard work the economist informed the company that the demand for the firm's novel (Θ_x) is given by the following equation:
$$\Theta_x = 12000 - 5000 P_x + 5I + 500 P_c$$

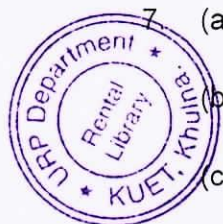
Assume that the initial values of P_x , I and P_c are \$5, \$10,000 and \$6 respectively.

Using the above information, do the followings on behalf of the Company Manager –

 - i) Determine what effect a price increase would have on total revenue.
 - ii) Evaluate how sale of the novels would change during a period of rising incomes.
 - iii) Assess the probable impact if competing publishers would raise their prices.
4. (a) Define markets. Write down the forms of market. (05)
(b) Define short-run. Write two conditions for equilibrium in perfect competition. (10)
(c) In perfect competition, show the point where the firm makes loss but still not shut down. (20)

Section – B

5. (a) What is GDP? Why do economists use real GDP rather than nominal GDP to gauge economic well-being? (10)
- (b) In the year 2001, the economy produces 100 loaves of bread that are sold for \$2 each. In the year 2002, the economy produces 200 loaves of bread that are sold for \$3 each. Calculate nominal GDP, Real GDP and the GDP deflator for each year using 2001 as base year. By what percentage does each of these three statistics rise from one year to next? (10)
- (c) What are the various problems of calculating national income? (15)
6. (a) What is inflation? What are the main causes of inflation? (10)
- (b) How can you suggest the government to control inflation by monetary measure? (15)
- (c) Distinguish between demand-pull inflation and cost-push inflation. (10)
7. (a) What is national savings, private savings and public savings? How are these three variables related? (10)
- (b) What is government budget deficit? How does it affect interest rates, investment, and economic growth? (10)
- (c) What is productivity? List and describe four determinants of productivity. (15)
8. (a) What is the role of Financial system? Mention two markets of the Financial system of Bangladesh. (15)
- (b) "Higher saving leads to a higher standard of living." Explain. (05)
- (c) Explain the factors of production. (15)
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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2016
Math 1217
(Mathematics - II)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 a. Define continuity of a function at a point. A function $f(x)$ is defined as follows: 15

$$f(x) = \begin{cases} x^2 & \text{when } x \leq 0 \\ x & \text{when } 0 < x < 1 \\ \frac{1}{x} & \text{when } x \geq 1 \end{cases}$$
 Discuss the continuity and differentiability of the function $f(x)$ at $x=0$
- b. If $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$, then find $\frac{dy}{dx}$ 12
- c. Find the n-th derivative of $\frac{1}{x^2 - a^2}$ 08
- 2 a. Show that $f(x) = \sin x + \sin x \cos x$ has a maximum value at $x = \frac{\pi}{3}$. Also find the maximum value. 10
- b. State Leibnitz's theorem. If $y = \sin(m \sin^{-1} x)$, then prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (m^2 - n^2)y_n = 0$ 13
- c. Form a differential equation from the relation $y = Ae^{3x} + Be^{-2x} + \cos 5x$, where A and B are constants. Finally write down order and degree of the differential equation. 12
- 3 Solve any three of the following differential equations: 35
 - a. $(\sqrt{1 + x^2 + y^2 + x^2 y^2}) dx + xy dy = 0$
 - b. $(x + y)dy + (x - y)dx = 0$
 - c. $(1 + e^{x/y})dx + e^{x/y}(1 - x/y)dy = 0$
 - d. $(2x - 2y + 5)dy - (x - y + 3)dx = 0$
- 4 Determine the general solution of any three of the followings: 35
 - a. $(D^3 - 2D^2 + 5D)y = 10 + 5 \cos 2x$
 - b. $(D^2 - 4D + 4)y = x^2 e^{2x}$
 - c. $(D^2 + 1)y = \sec x$
 - d. $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - 12y = (x - 1)e^{2x}$

Section – B

5 Integrate any three of the followings:

35

a. $\int (2x + 3)\sqrt{4x^2 + 8x + 5} \, dx$

b. $\int e^x \left\{ \frac{x}{(1+x)^2} \right\} dx$

c. $\int \frac{dx}{(1-x)\sqrt{1+x^2}}$

d. $\int \frac{7x - 9}{x^2 - 2x + 35} dx$



6 Evaluate any three of the followings:

35

a. $\int_0^\pi x \log \sin x \, dx$

b. $\int_0^1 \frac{\log(1+x)}{1+x^2} dx$

c. $\int_0^{\pi/4} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$

d. $\int_0^{\pi/2} \frac{dx}{1 + \sqrt{\tan x}}$

7 a. Obtain a reduction formula for $\int \cos^m x \cos nx \, dx$ and hence find the value of $\int_0^{\pi/2} \cos^3 x \cos 2x \, dx$

15

b. Find the area bounded by the curve $x^2 = 4ay$ and the line $y = x$.

10

c. Represent the following series as a definite integral and hence evaluate:

10

$$\text{Lt}_{n \rightarrow \infty} \left[\frac{n+2}{n^2+1} + \frac{n+4}{n^2+4} + \frac{n+6}{n^2+9} + \dots + \frac{3}{2n} \right]$$

8 a. If $u_n = \int_0^1 x^n \tan^{-1} x \, dx$, then prove that $(n+1)u_n + (n-1)u_{n-2} = \frac{\pi}{2} - \frac{1}{n}$

13

b. Find the area bounded by the curve $x^{2/3} + y^{2/3} = a^{2/3}$

10

c. Evaluate $\int_0^1 \cot^{-1}(1-x+x^2) \, dx$

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Khulna University of Engineering & Technology
 Department of Urban and Regional Planning
 BURP 1st Year 2nd Term Regular Examination, 2016
URP 1211
 (Urban Planning Principles)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
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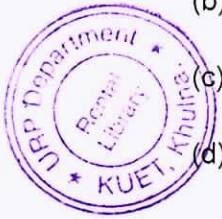
Section – A

1. (a) Define urban planning by Keeble? Explain the scopes of urban planning with examples. (15)
 (b) Draw a schematic diagram to represent different types of New Towns. (05)
 (c) Elucidate the role of New Town Policy in regional balance. (05)
 (d) Write down the salient characteristics of Satellite Town. (10)
2. (a) Discuss the neighbourhood unit concept by Clarence A. Perry with necessary schematic diagram. (12)
 (b) "Too high density is not acceptable for urban planning" - why? (05)
 (c) A Real Estate company is planning to build a two-story building on a rectangular plot that has 100 feet of street frontage and 200 feet of depth. The first story measures 50 feet by 200 feet and the second story measures 50 feet by 200 feet. Calculate FAR of the building. (08)
 (d) How intensity of a residential area is expressed? Explain with examples? (10)
3. (a) Define open space according to M. Jeanne Davis. (05)
 (b) Discuss different types of Regional Open space with examples. (08)
 (c) Write down in your own words the causes, consequences and remedies of disappearance of open space in the cities and towns of Bangladesh with examples. (10)
 (d) Suppose, you are working as a transport planner for Khulna Railway Station. What are the principles you should follow for a good railway design? (12)
4. (a) What factors a transport planner should consider for developing a new Bus Terminal in Khulna City? (13)
 (b) What are the advantages and disadvantages of Nearside placement of bus stop? (10)
 (c) Give a diagram to show maximum walking distance of a pedestrian to station facilities from station entrance. (06)
 (d) How critical dimensions affect layout design of a bus stop? (06)

Section – B

5. (a) Make a comparative analysis on radial, grid and irregular road pattern of urban area. Which one will you prefer for designing a commercial area? Why? (10)
 (b) "Indicators of land value for residential, industrial and commercial use are similar" – Justify the statement with appropriate example. (10)
 (c) Write your concept on Rank Size Rule. Is it applicable in your country? Justify your answer. (05)
 (d) Clarify any two of the following issues: (10)
 (i) Urban system (ii) Urban Pyramid (iii) Primate city (iv) Central place theory

6. (a) Make a comparative analysis between rooted and linked industry. Explain the relationship between these industries with proper example. (08)
- (b) "Industrial park includes all crucial facilities for industrial development" – write about the service facilities available in any industrial park with necessary example. (12)
- (c) "Eco-industrial park ensures maximum utilization of available resources" – explain. Use proper example. (08)
- (d) Industrial development can be encouraged through persuasion – how? Provide an example. (07)
7. (a) "Socio-economic data analysis is a vital part of commercial area selection" – why? Explain with appropriate example. (10)
- (b) Make a list of spatial considerations necessary for the selection of a land for commercial purpose. Provide an example. (07)
- (c) "Vehicular mobility gradually increases and land access decreases from local roads to freeways" – explain. Use necessary diagram. (08)
- (d) Provide a short overview of road classification provided by Local Government Engineering Department (LGED) for your country. (10)
8. (a) Narrate some common characteristics of town centre as a vibrant gathering place of the community. (05)
- (b) "Ensuring proper balance between vehicular and pedestrian movement is a vital challenge of town centre design" – how this problem can be solved? Explain. (15)
- (c) Explain any three of the following principles for successful town centre design: (15)
- (i) Balance flexibility with a long term vision
 - (ii) Integrate multiple uses
 - (iii) Capture the benefit that density offers
 - (iv) Invest for sustainability
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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2016
URP 1251
(Introduction to Architectural History)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Signify the "History of Architecture" as a student of planning. Write down the factors which are influential in the History of Architecture. (05)
(b) Analyse the characteristics of ancient Egyptian Architecture with relevant sketches. (15)
(c) "Classical Greek Architecture is enhanced by its innovative solutions"-explain the statement with necessary sketches. (15)
2. (a) Explain the architectural characteristics of classical Roman period with necessary sketches. (15)
(b) "Hagia Sophia is a remarkable example of Byzantine period"-evaluate. (10)
(c) Describe St. Peter's Basilica (Rome) with relevant sketches. (10)
3. (a) "Renaissance style places emphasis on symmetry, proportion and geometry with the architecture of classical antiquity"- prove the statement with proper sketches. (20)
(b) Assess the architectural characteristics of Gothic period. (10)
(c) Identify the distinctive characteristics of Baroque Architecture. (05)
4. (a) Evaluate the background factors of modern architectural movements. (10)
(b) "Form Follows Function"-explain the statement in terms of modern architecture. (15)
(c) Compare town planning approaches of modern period. (10)

Section – B

5. (a) "Wind, water and clay shape the settlement pattern of Bengal"-explain. (10)
(b) Write down your understanding about "Deltaic Pavilion". (15)
(c) Identify the contributions of architect Muzharul Islam. (10)
6. (a) "Influence of Buddhism flourished as a new taste in ancient Bengal"-how will you justify? (15)
(b) Philosophically interpret the Buddhist Vihara with necessary sketches. (10)
(c) Assess the importance of clay as an important construction material of Bengal. (10)
7. (a) Classify Hindu temples and give examples. (15)
(b) Characterise a typical Hindu temple with necessary sketches. (15)
(c) Make a comparison between Gupta dynasty and Pala dynasty. (05)
8. (a) Identify the features of Bungalow which are influential for its development outside urban area. (10)
(b) Figure out the basic layout plan of Feudal Palaces with sketches. (15)
(c) Find out the significant features of Shait Gumbad mosque. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2016
URP 1281
(Surveying and Cartography)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.



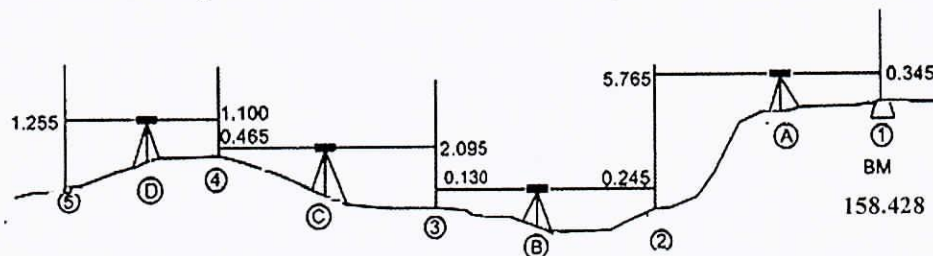
Section – A

1. (a) What is the difference between surveying and leveling? (05)
- (b) Where chain survey is recommended? Explain with necessary example. (10)
- (c) The following consecutive readings were taken with a level and a 4-meter leveling staff on a continuously sloping ground at common intervals of 30 m: 0.855 m (on A), 1.545 m, 2.335 m, 3.115 m, 3.825 m, 0.0455 m, 1.380 m, 2.055 m, 3.455 m, 0.585 m, 1.015 m, 1.850 m, 2.755 m and 3.845 m (on B). The RL of A was 380.500. Make entries in a level book and apply the usual checks. Determine the slope of AB. The change points were taken at 120 m and 240 m in chainage. (20)

2. (a) What is your understanding about isogonic and agonic lines? Mention the causes responsible for the variation of magnetic declination. (10)
- (b) Explain the Bowditch's graphical method of traverse adjustment with necessary diagram. (08)
- (c) The following are the bearings observed in traversing, with a compass, an area where local attraction was suspected. Calculate the interior angles of the traverse and correct them if necessary. (12)

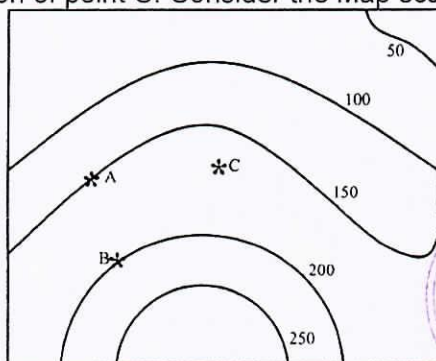
Line	FB	BB
AB	191°45'	13°0'
BC	39°30'	222°30'
CD	22°15'	200°30'
DE	242°45'	62°45'
EA	330°15'	147°45'

- (d) On an old map a line was drawn to a magnetic bearing of 320°30', when the declination was 3°30' W. Find the present bearing of the line, if the declination is 4°15' E. (05)
3. (a) What are the methods of plane tabling? Mention the field procedure of setting up plane table over a station. (07)
- (b) What do you mean by orientation in plane table surveying? How is it done? Explain with necessary sketches. (10)
- (c) The following figure is the outcome of levelling survey. Calculate the elevations using rise and fall method and check your result. (18)



4. (a) Explain the comparison between Collimation system and Rise and Fall system. (10)
- (b) Explain the characteristics of contour with necessary sketches. (10)

- (c) From the following contour map find the slope between points A and B. Also calculate the elevation of point C. Consider the Map scale: 1 inch = 1,000 ft. (10)



- (d) Define the terms "contour interval" and "horizontal equivalent". (05)

Section – B

5. (a) Maps allow many types of relationships to be formed. Give examples of the following: (10)
- Relationships among locations with no attributes.
 - Relationship among various attributes at the same point.
 - Relationship among different locations of the same attribute.
 - Relationships among locations of combined/derived attributes of given distributions.
 - Relationships among attributes with no location.
- (b) Mention the type of outputs one can get from cartography. (06)
- (c) Convert the Proportionate Scale (R.F) 1: 3960 into text scale, in inch and mile. (04)
- (d) "A map is not a photograph" - explain the statement with proper example. (10)
- (e) Classify maps based on scale with proper example. (05)

6. (a) Briefly explain the different types of thematic map with example. (20)
- (b) Consider the following scenario in a district with 4 zones: (15)

Zone	Population (in thousands)	Male to Female Ratio
A	1254	0.65
B	6250	0.45
C	4632	0.50
D	7412	0.55

Outline the process of creating a suitable map (type and design) that shows the Male to Female Ratio.

7. (a) Why is BTM projection system used even though there is a global projection system (UTM)? (05)
- (b) Explain why different datums give different pointers for the same location. (05)
- (c) Explain with example, the different types of distortion with relation to the basic map projection. (10)
- (d) What is geoid? Explain why it is modeled from gravity data. (10)
- (e) Differentiate between Ellipsoid and Geoid. (05)
8. (a) For the Earth, $a = 1750$ km; $b = 2555$ km. Calculate the flattening ratio. (05)
- (b) With the help of appropriate figures explain the various types of Coordinate Systems. (10)
- (c) Explain RMS error with relation to Geo-referencing. Why does it need to be less than one? (10)
- (d) Describe 5 types of Transformation for Geo-referencing. (10)